Project Reference Guide

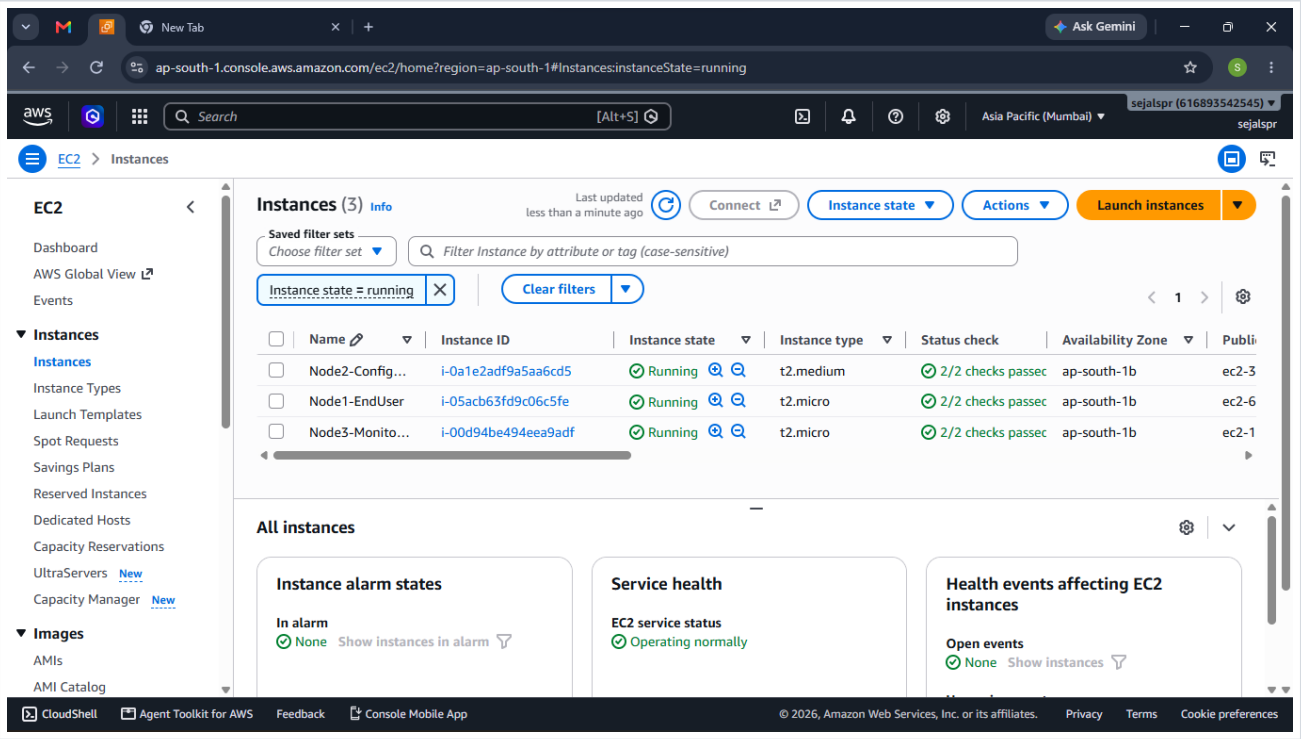
# Automated CI/CD Pipeline & 3-Node AWS Architecture

Swiggy Clone deployed with Apache, Jenkins, CloudWatch and SNS on AWS EC2 (Ubuntu).
Target roles: Cloud Engineer / DevOps Engineer / System Administrator.

## 1. Infrastructure Provisioning (AWS EC2)

Provisioned three separate Ubuntu instances to separate workloads cleanly:

- **Node 1 (Node1-EndUser):** Dedicated Apache web server.
- **Node 2 (Node2-Configuring):** Dedicated Docker and Jenkins host.
- **Node 3 (Node3-Monitoring):** Dedicated observability and monitoring node.

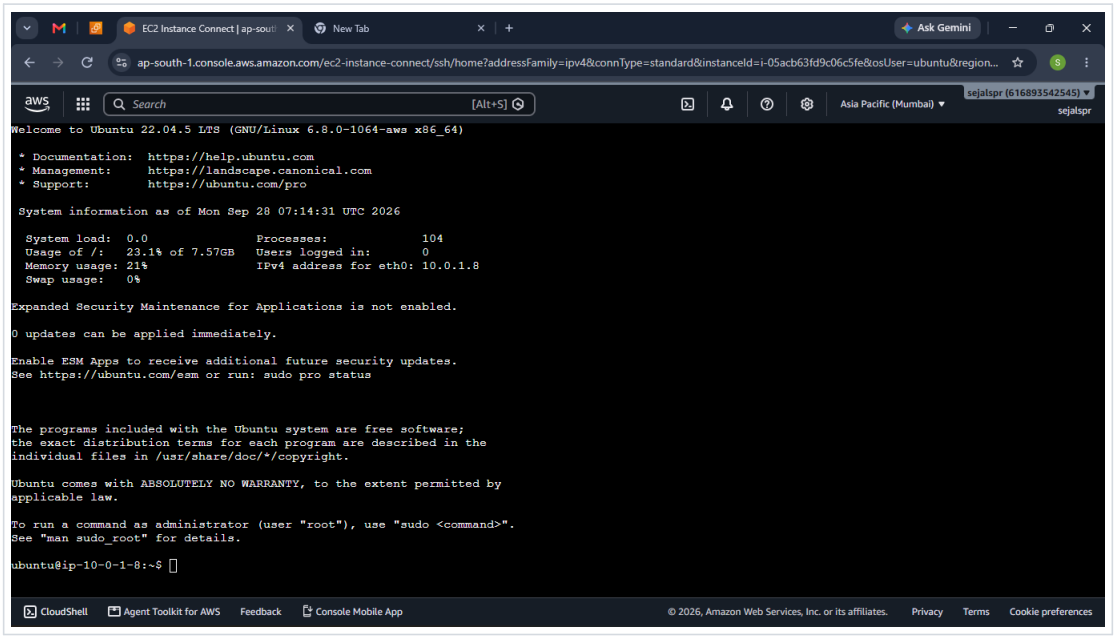



*Three EC2 instances running in ap-south-1 (Mumbai)*

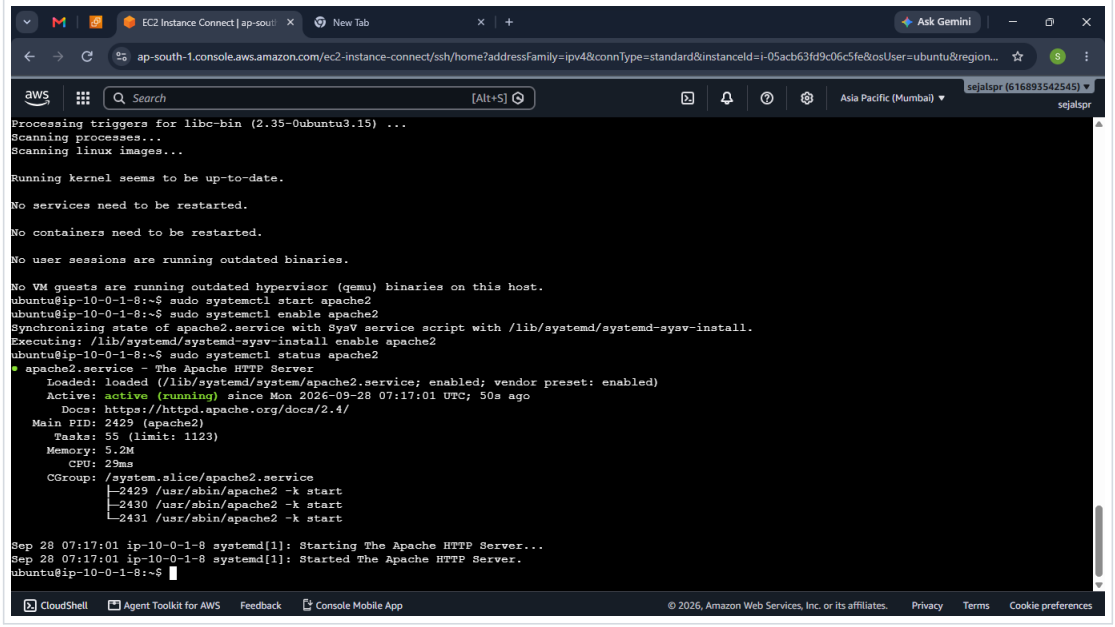
## 2. Node 1: Apache Web Server Setup & Deployment

Connected to Node 1 via EC2 Instance Connect and ran the following commands to install and start Apache:

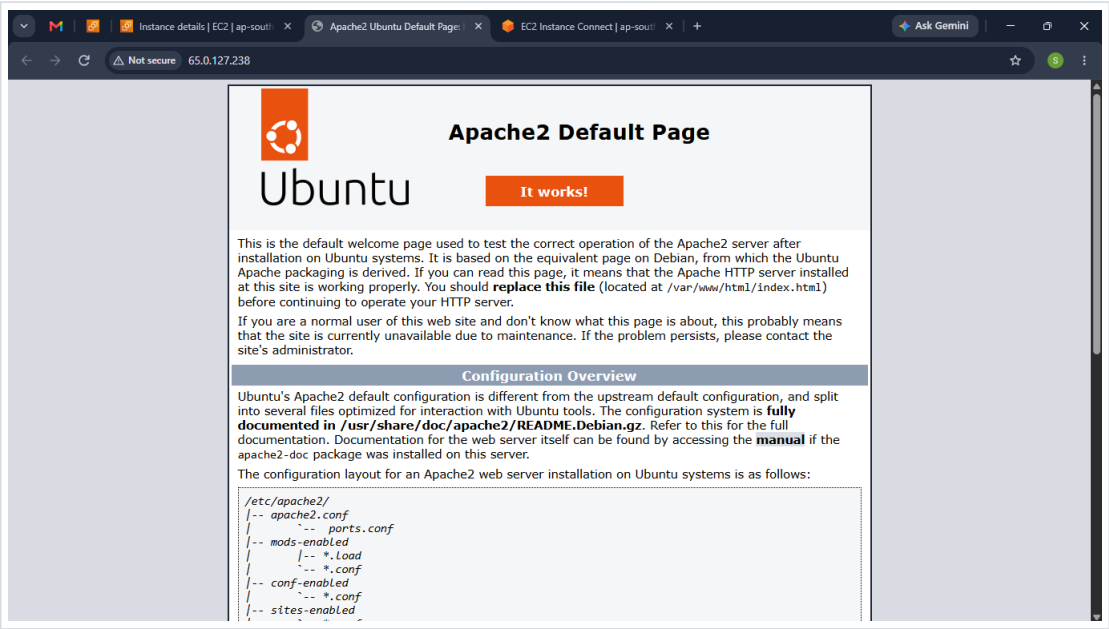
```
sudo apt update -y
sudo apt install apache2 -y
sudo systemctl start apache2
sudo systemctl enable apache2
sudo systemctl status apache2
```




*Connected to Node 1 via EC2 Instance Connect*

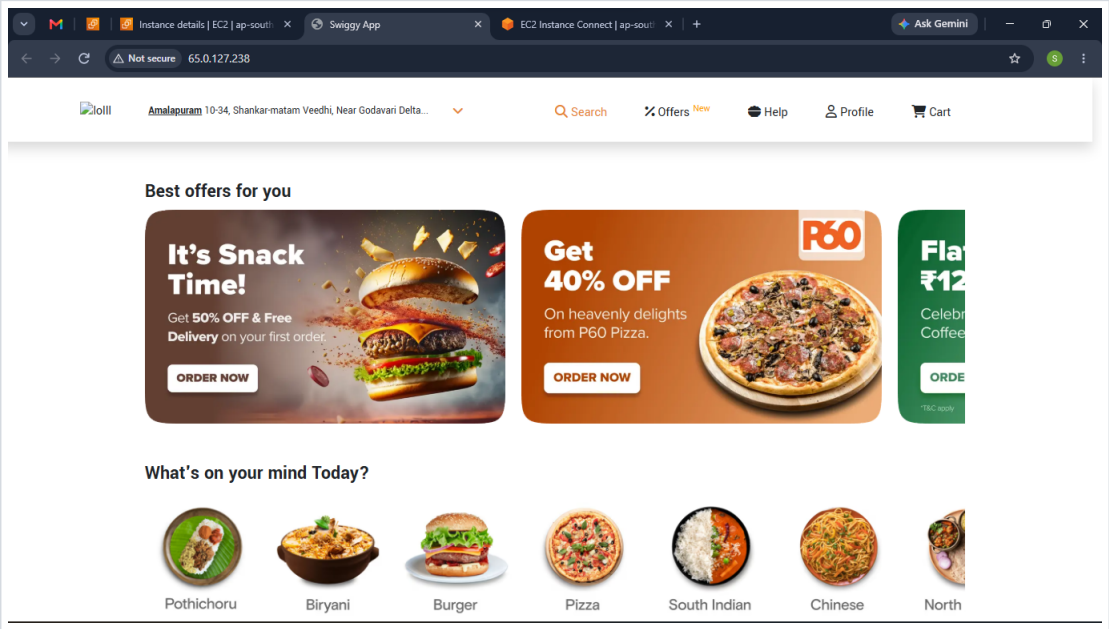



*apache2 service active (running)*




*Apache2 default page served*

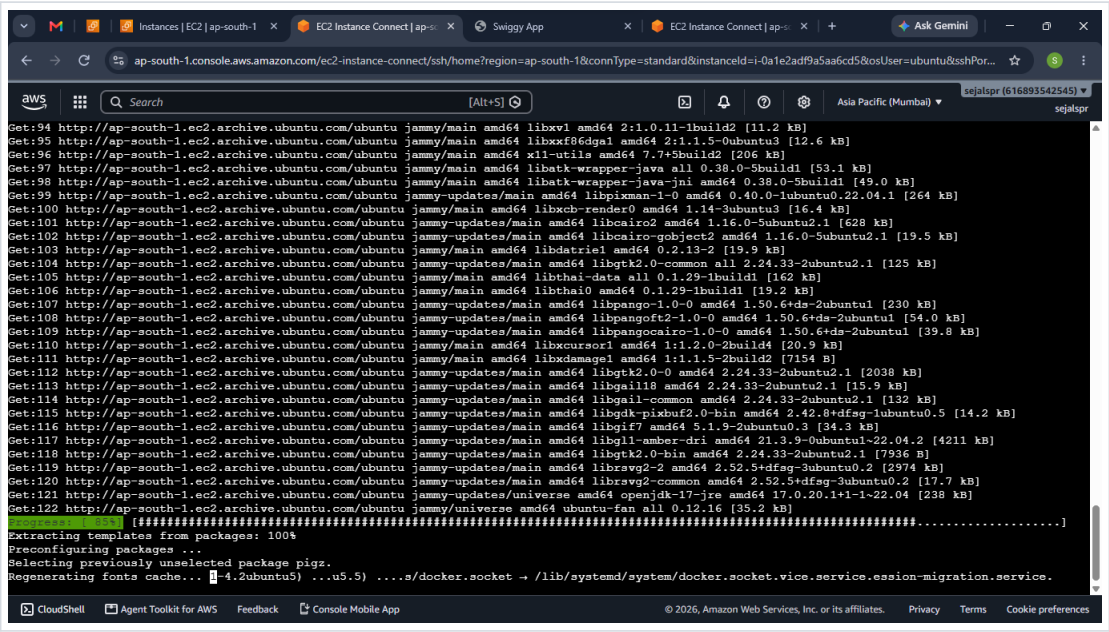
**Result:** Served the default Apache page and later deployed the live Swiggy React application build to /var/www/html/.



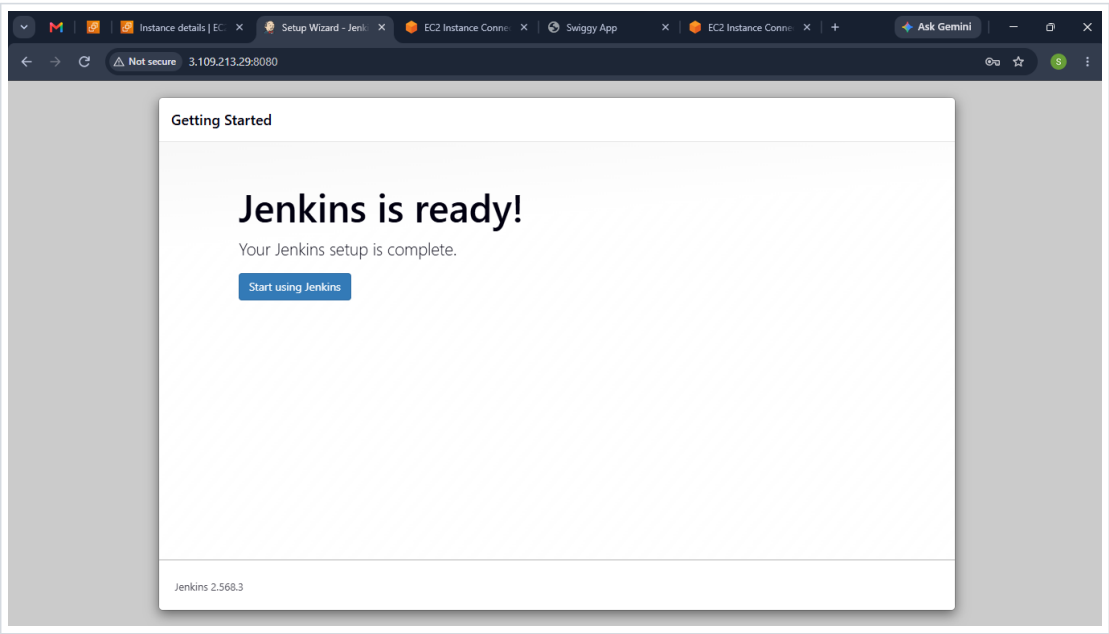

*Swiggy React clone running live*

## 3. Node 2: Jenkins CI/CD Pipeline Configuration

- **Environment setup:** Installed Docker, Git and Java runtimes on Node 2.
- **Jenkins installation:** Deployed and initialized the Jenkins server dashboard.



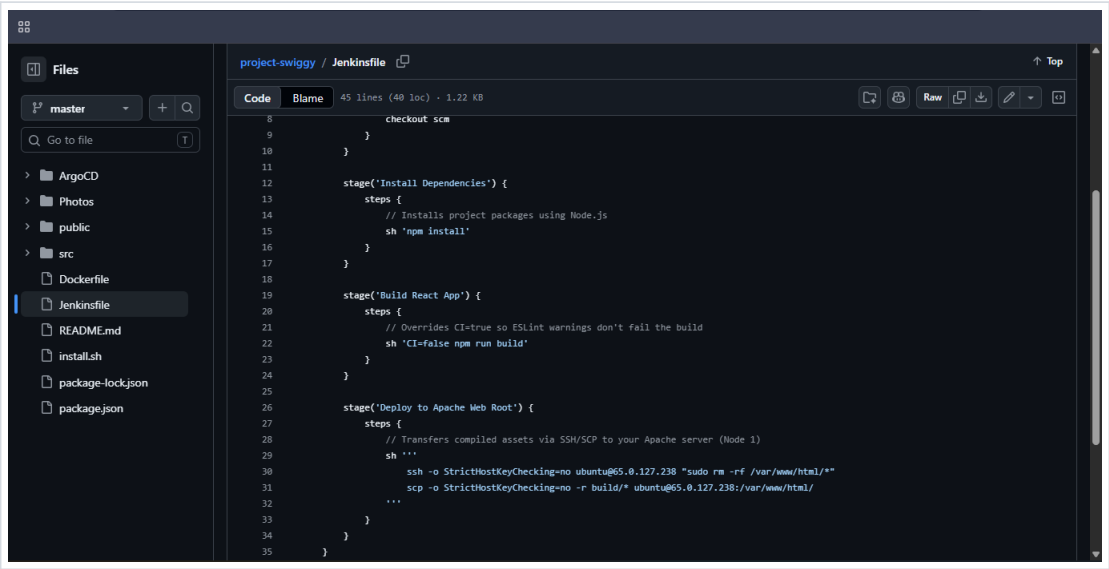

*Installing Docker, Git and Java on Node 2*




*Jenkins is ready*

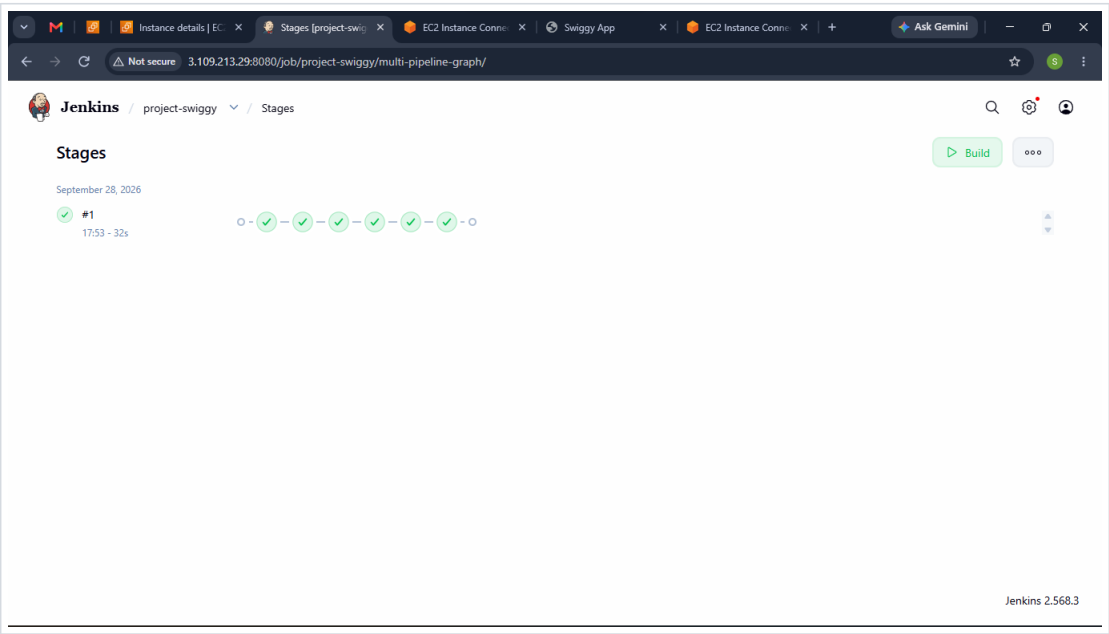
**Pipeline script (Jenkinsfile):** Declarative pipeline with these key stages:

- **Checkout SCM:** Pulls the latest code from the project-swiggy GitHub repository.
- **Install Dependencies:** Runs npm install.
- **Build React App:** Compiles the application using CI=false npm run build.
- **Deploy to Apache Web Root:** Uses SSH/SCP to push compiled assets (build/*) to Node 1.




*Jenkinsfile in the project-swiggy repository*

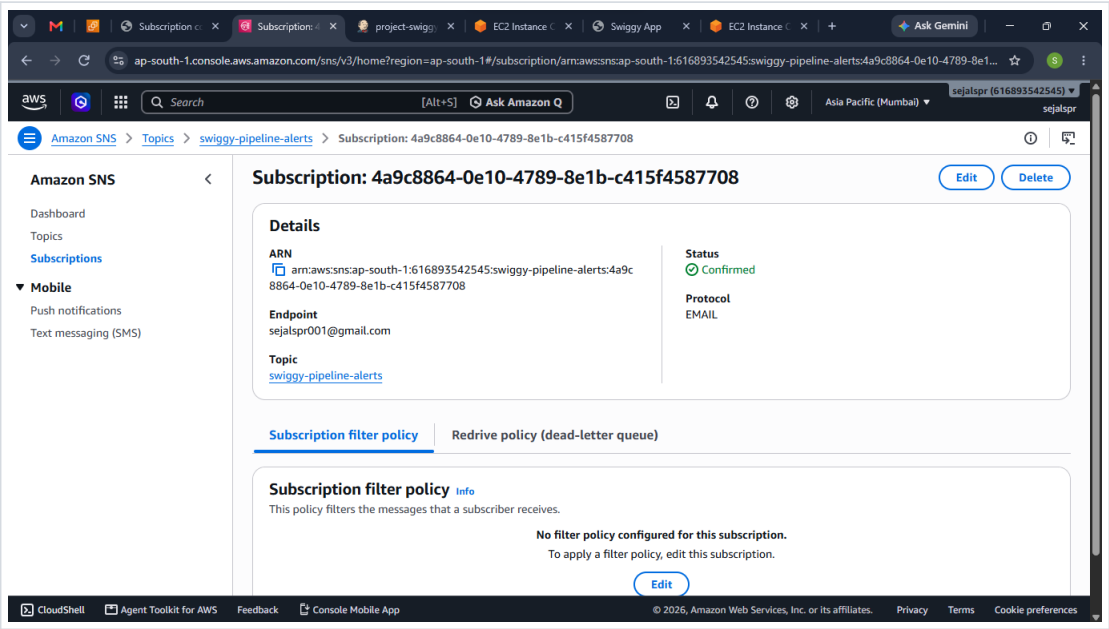
**Execution:** Successfully triggered automated pipeline builds (Build #1 completed successfully in 32 seconds).




*Jenkins pipeline stages: Build #1 successful*

## 4. Node 3 & System Monitoring (CloudWatch & SNS)

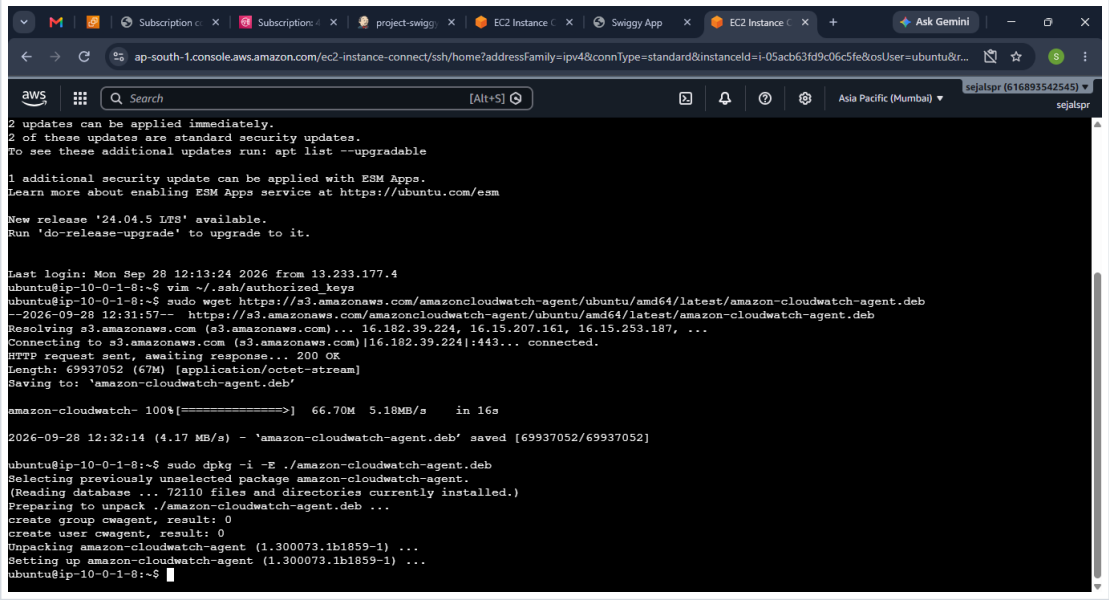
- **SNS alert setup:** Created a standard Amazon SNS topic (swiggy-pipeline-alerts) with a confirmed email subscription endpoint.



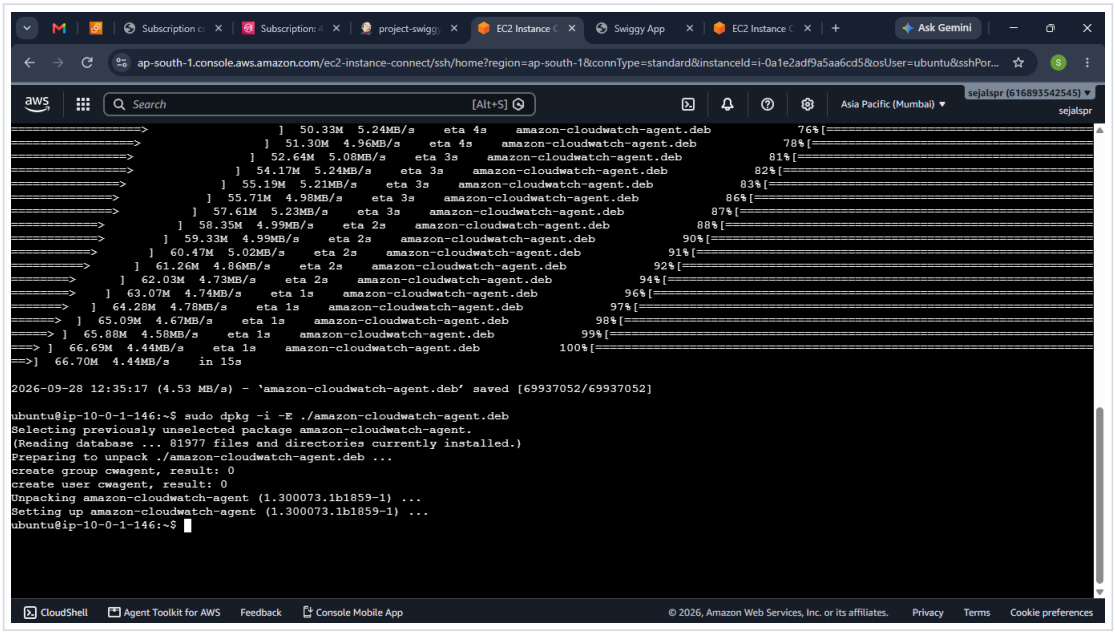

*SNS subscription status: Confirmed*

**CloudWatch agent installation:** Downloaded and installed the CloudWatch agent package across the infrastructure nodes:

```
sudo wget https://s3.amazonaws.com/amazoncloudwatch-agent/ubuntu/amd64/latest/
     amazon-cloudwatch-agent.deb
sudo dpkg -i -E ./amazon-cloudwatch-agent.deb
```

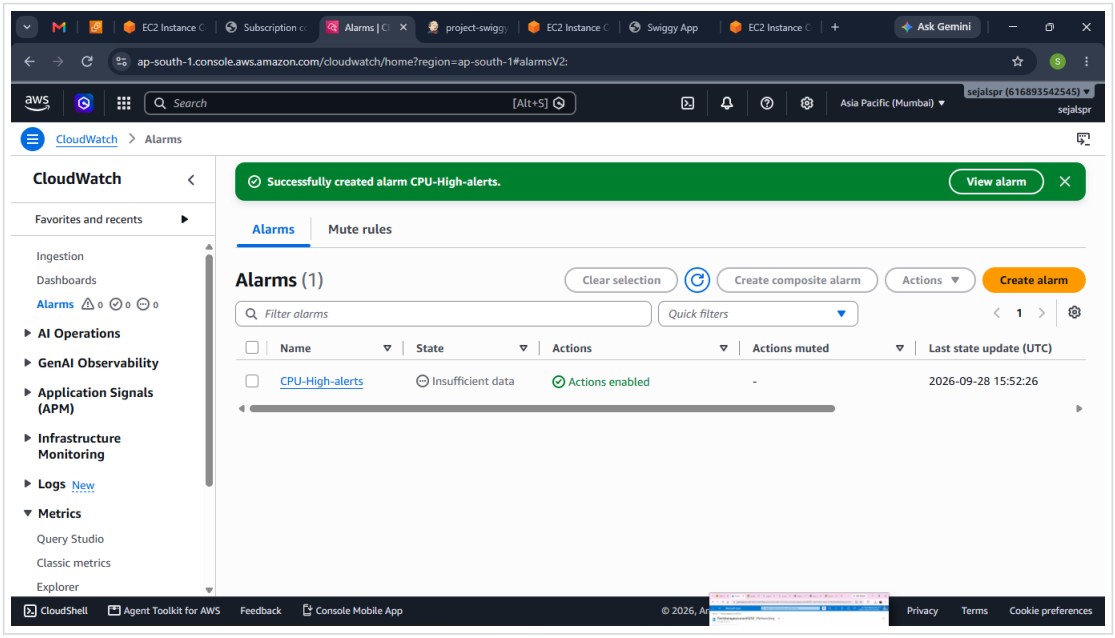



*CloudWatch agent installed on Node 1*

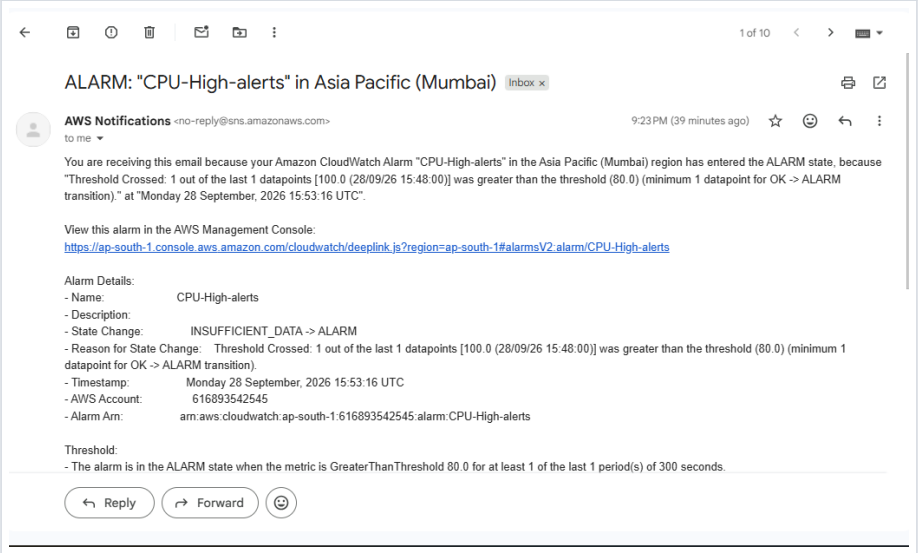



*CloudWatch agent installed on Node 2*

**Alarms & observability:** Created automated health alarms (such as CPU-High-alerts) to track performance and trigger alerts.




*CPU-High-alerts alarm created in CloudWatch*




*Alert email delivered through SNS*

## 5. Project Outcome

- **Node 1:** Hosting the live, automated Swiggy React application via Apache.
- **Node 2:** Running the Dockerized Jenkins CI/CD server that automatically builds and deploys code updates.
- **Node 3:** Monitoring system performance with CloudWatch and keeping infrastructure safe with SNS alerts.